

# Integration & Applications



## Basic antiderivatives (the power rule)

- 1 Find  $\int x^4 dx$ .
  - 2 Find  $\int 3x^2 dx$ .
  - 3 Find  $\int \sqrt{x} dx$ .
  - 4 Find  $\int \frac{1}{x^3} dx$ .
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## Antiderivatives of polynomials

- 5 Find  $\int (4x^3 - 6x + 5) dx$ .
  - 6 Find  $\int (2x + 7) dx$ .
  - 7 Find  $\int (x^2 - 4x + 3) dx$ .
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## Antiderivatives of trigonometric and exponential functions

- 8 Find  $\int \cos x dx$ .
  - 9 Find  $\int \sin x dx$ .
  - 10 Find  $\int e^x dx$ .
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## Evaluating definite integrals

- 11 Evaluate  $\int_0^2 3x^2 dx$ .
  - 12 Evaluate  $\int_1^3 2x dx$ .
  - 13 Evaluate  $\int_0^4 x dx$ .
  - 14 Evaluate  $\int_{-1}^2 3x^2 dx$ .
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## Definite integrals with polynomial integrands

15 Evaluate  $\int_0^2 (x^2 + 1) dx$ .

16 Evaluate  $\int_1^3 (2x^2 - 3x + 1) dx$ .

17 Evaluate  $\int_0^1 (6x^2 - 4x + 2) dx$ .

18 Evaluate  $\int_{-2}^1 (x^2 - x) dx$ .

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## Definite integrals of trigonometric and exponential functions

19 Evaluate  $\int_0^{\pi/2} \cos x dx$ .

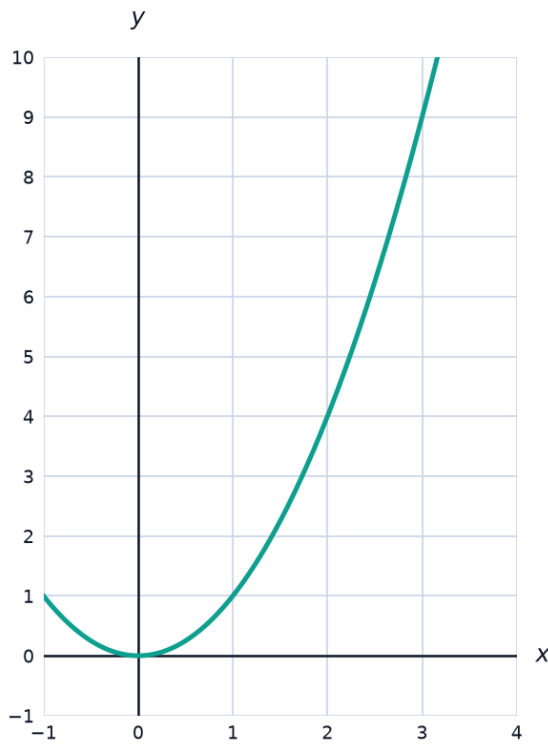
20 Evaluate  $\int_0^{\pi} \sin x dx$ .

21 Evaluate  $\int_0^1 e^x dx$ .

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## Area under a curve

22 Find the area under  $y = x^2$  from  $x = 0$  to  $x = 3$ , using a definite integral.



23 Find the area under  $y = 4 - x^2$  from  $x = -2$  to  $x = 2$ .

- 24 Find the area under  $y = 3\sqrt{x}$  from  $x = 0$  to  $x = 4$ .
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### Signed area and regions below the axis

- 25 Evaluate  $\int_0^2 (x - 2) dx$  and explain what the sign of the result means.
- 26 Find the (positive) area between  $y = x - 2$  and the  $x$ -axis, from  $x = 0$  to  $x = 2$ .
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### Properties of definite integrals

- 27 Given  $\int_1^4 f(x) dx = 10$ , find  $\int_1^4 3f(x) dx$ .
- 28 Given  $\int_0^2 f(x) dx = 5$  and  $\int_2^5 f(x) dx = 7$ , find  $\int_0^5 f(x) dx$ .
- 29 Given  $\int_0^3 f(x) dx = 8$ , find  $\int_3^0 f(x) dx$ .
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### Finding a function from its derivative

- 30 A function satisfies  $f'(x) = 6x^2 - 4$  and  $f(1) = 5$ . Find  $f(x)$ .
- 31 A function satisfies  $f'(x) = 3x^2 + 2x$  and  $f(0) = 1$ . Find  $f(x)$ .
- 32 A function satisfies  $f'(x) = 4x - 1$  and  $f(2) = 10$ . Find  $f(x)$ .
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### Average value of a function

- 33 Find the average value of  $f(x) = x^2$  on  $[0, 3]$ , using  $\text{avg} = \frac{1}{b-a} \int_a^b f(x) dx$ .
- 34 Find the average value of  $f(x) = 2x + 1$  on  $[1, 5]$ .
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### Word problems: integration in context

- 35 This question has two parts. A particle moves along a line with velocity  $v(t) = 6t - 3t^2$  (m/s) for  $0 \leq t \leq 4$ .
- a Find the particle's displacement from  $t = 0$  to  $t = 2$ , using  $\int_0^2 v(t) dt$ .
  - b Find the particle's displacement from  $t = 0$  to  $t = 4$ , and explain why it is not simply twice the part (a) answer.
- 36 This question has two parts. Water flows into a tank at a rate of  $r(t) = 20 - 2t$  liters per minute, for  $0 \leq t \leq 10$ .
- a Write a definite integral for the total volume of water that flows in during the first 5 minutes.
  - b Evaluate the integral to find the volume.
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### MCQ — Basic antiderivatives (the power rule)

- 37 Find  $\int x^6 dx$ .
- a.  $\frac{x^6}{7} + C$
  - b.  $7x^5 + C$
  - c.  $\frac{x^7}{6} + C$
  - d.  $\frac{x^7}{7} + C$
- 38 Find  $\int 5x^3 dx$ .
- a.  $5x^4 + C$
  - b.  $\frac{5x^4}{4} + C$
  - c.  $\frac{5x^4}{3} + C$
  - d.  $15x^2 + C$
- 39 Find  $\int \sqrt[3]{x} dx$ .
- a.  $\frac{3}{2}x^{2/3} + C$
  - b.  $\frac{4}{3}x^{4/3} + C$
  - c.  $\frac{3}{4}x^{4/3} + C$
  - d.  $\frac{1}{3}x^{-2/3} + C$

40 Find  $\int \frac{1}{x^4} dx$ .

- a.  $\frac{1}{3x^3} + C$
- b.  $-\frac{1}{4x^3} + C$
- c.  $-\frac{1}{3x^3} + C$
- d.  $-\frac{4}{x^5} + C$

41 Find  $\int 7 dx$ .

- a.  $C$
- b.  $7x + C$
- c.  $7 + C$
- d.  $\frac{7x^2}{2} + C$

42 Find  $\int \frac{1}{\sqrt{x}} dx$ .

- a.  $2\sqrt{x} + C$
- b.  $\frac{1}{2}\sqrt{x} + C$
- c.  $-2\sqrt{x} + C$
- d.  $\sqrt{x} + C$

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### MCQ — Antiderivatives of polynomials

43 Find  $\int (6x^2 - 8x + 3) dx$ .

- a.  $2x^3 - 4x^2 + 3x + C$
- b.  $2x^3 - 4x^2 + C$
- c.  $2x^3 - 8x^2 + 3x + C$
- d.  $12x - 8 + C$

- 44 Find  $\int (5x + 9) dx$ .
- a.  $\frac{5x^2}{2} + C$
  - b.  $5x^2 + 9x + C$
  - c.  $\frac{5x^2}{2} + 9x + C$
  - d.  $\frac{5x^2}{2} + 9 + C$
- 45 Find  $\int (x^2 - 6x + 5) dx$ .
- a.  $\frac{x^3}{3} - 3x^2 + 5x + C$
  - b.  $\frac{x^3}{3} - 6x^2 + 5x + C$
  - c.  $\frac{x^3}{3} - 3x^2 + C$
  - d.  $x^3 - 3x^2 + 5x + C$

- 46 Find  $\int (3x^3 - 2x) dx$ .
- a.  $3x^4 - x^2 + C$
  - b.  $\frac{3x^4}{4} - x^2 + C$
  - c.  $\frac{3x^4}{4} - 2x^2 + C$
  - d.  $\frac{x^4}{4} - x^2 + C$

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### MCQ — Antiderivatives of trigonometric and exponential functions

- 47 Find  $\int 2\cos x dx$ .
- a.  $2\sin x + C$
  - b.  $-2\sin x + C$
  - c.  $2\cos x + C$
  - d.  $\sin(2x) + C$

- 48 Find  $\int 3\sin x \, dx$ .
- a.  $3\sin x + C$
  - b.  $-3\sin x + C$
  - c.  $3\cos x + C$
  - d.  $-3\cos x + C$
- 49 Find  $\int (e^x + 2) \, dx$ .
- a.  $e^x + 2 + C$
  - b.  $e^x + 2x + C$
  - c.  $xe^x + 2x + C$
  - d.  $e^x + C$
- 50 Find  $\int (\sin x + \cos x) \, dx$ .
- a.  $-\cos x - \sin x + C$
  - b.  $\cos x + \sin x + C$
  - c.  $\cos x - \sin x + C$
  - d.  $-\cos x + \sin x + C$

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### MCQ — Evaluating definite integrals

- 51 Evaluate  $\int_0^3 2x \, dx$ .
- a. 9
  - b. 6
  - c. 3
  - d. 18

52 Evaluate  $\int_1^4 3x^2 dx$ .

a. 21

b. 48

c. 64

d. 63

53 Evaluate  $\int_0^5 x dx$ .

a. 5

b. 12.5

c. 25

d. 10

54 Evaluate  $\int_{-2}^1 4x^3 dx$ .

a. -17

b. -15

c. 15

d. 1

55 Evaluate  $\int_2^5 3 dx$ .

a. 3

b. 9

c. 6

d. 15

56 Evaluate  $\int_0^2 6x^2 dx$ .

- a. 16
- b. 12
- c. 24
- d. 8

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**MCQ — Definite integrals with polynomial integrands**

57 Evaluate  $\int_0^3 (x^2 + 2) dx$ .

- a. 9
- b. 15
- c. 11
- d. 21

58 Evaluate  $\int_1^4 (3x^2 - 2x + 1) dx$ .

- a. 50
- b. 52
- c. 63
- d. 51

59 Evaluate  $\int_0^2 (4x^2 - 2x + 3) dx$ .

- a.  $\frac{32}{3}$
- b.  $\frac{40}{3}$
- c.  $\frac{38}{3}$
- d. 14

60 Evaluate  $\int_{-1}^2 (x^2 - x) dx$ .

- a. 4.5
- b. 2.5
- c. 0.5
- d. 1.5

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### MCQ — Definite integrals of trigonometric and exponential functions

61 Evaluate  $\int_0^{\pi/2} \sin x dx$ .

- a.  $\frac{\pi}{2}$
- b.  $-1$
- c. 0
- d. 1

62 Evaluate  $\int_0^{\pi} \cos x dx$ .

- a.  $\pi$
- b.  $-2$
- c. 0
- d. 2

63 Evaluate  $\int_0^2 e^x dx$ .

- a.  $e^2 - 1$
- b.  $2e$
- c.  $e^2 + 1$
- d.  $e^2$

### MCQ — Area under a curve

64 Find the area under  $y = x^2$  from  $x = 0$  to  $x = 4$ , using a definite integral.

a.  $\frac{64}{3}$

b. 64

c. 16

d.  $\frac{16}{3}$

65 Find the area under  $y = 9 - x^2$  from  $x = -3$  to  $x = 3$ .

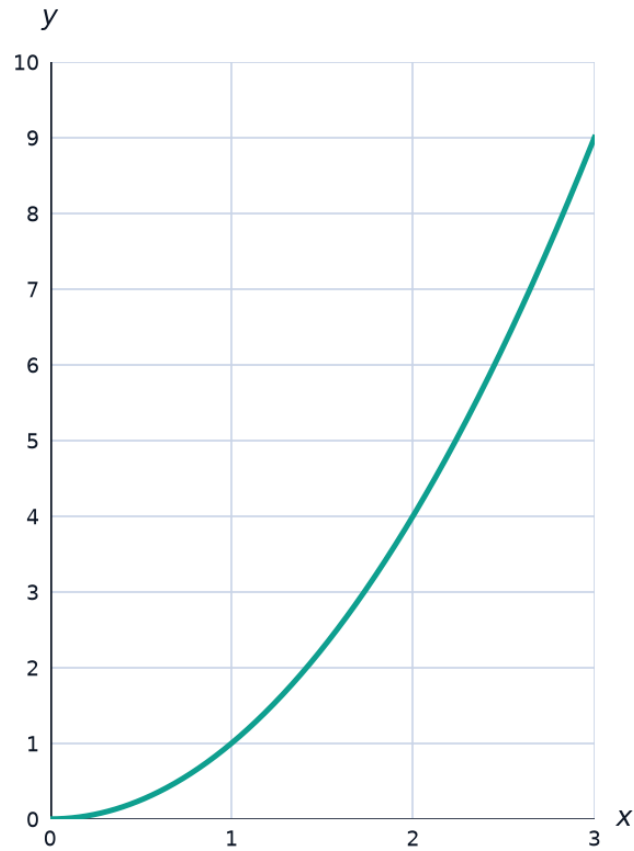
a. 27

b. 54

c. 36

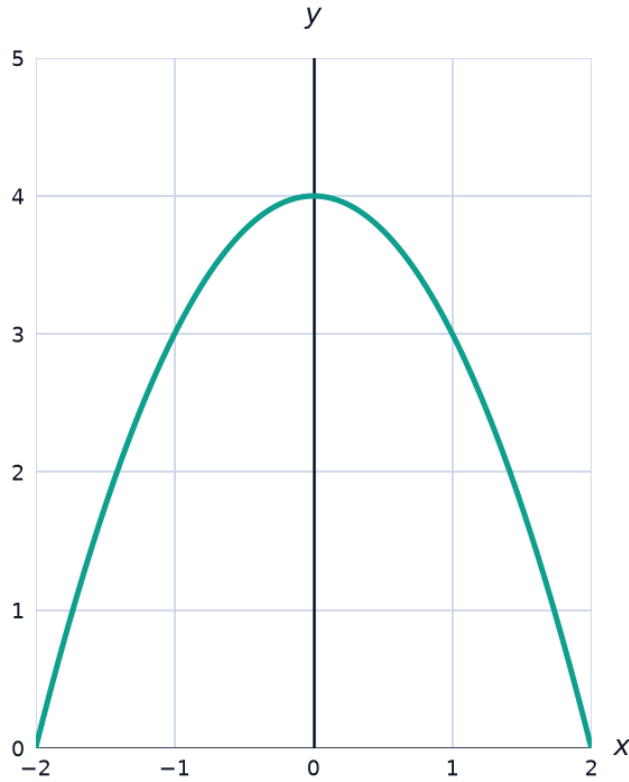
d. 18

- 66 The graph shows  $y = x^2$ . Find the area under the curve from  $x = 0$  to  $x = 3$ , using a definite integral.



- a. 9
- b. 18
- c. 3
- d. 27

- 67 The graph shows  $y = 4 - x^2$ . Find the area under the curve from  $x = -2$  to  $x = 2$ .



- a.  $\frac{16}{3}$
- b. 32
- c. 16
- d.  $\frac{32}{3}$

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### MCQ — Signed area and regions below the axis

- 68 Evaluate  $\int_0^3 (x - 3) dx$  and explain what the sign of the result means.
- a. 4.5; the region lies above the x-axis
  - b. -4.5; the region lies above the x-axis
  - c. 4.5; the region lies below the x-axis
  - d. -4.5; the region lies below the x-axis

- 69 Find the (positive) area between  $y = x - 3$  and the  $x$ -axis, from  $x = 0$  to  $x = 3$ .
- a. 9
  - b.  $-4.5$
  - c. 3
  - d. 4.5
- 70 Evaluate  $\int_0^4 (x - 4) dx$  and state whether the region is above or below the  $x$ -axis.
- a. 8; above
  - b.  $-8$ ; above
  - c. 8; below
  - d.  $-8$ ; below
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### MCQ — Properties of definite integrals

- 71 Given  $\int_2^6 f(x) dx = 12$ , find  $\int_2^6 4f(x) dx$ .
- a. 12
  - b. 16
  - c. 48
  - d. 3
- 72 Given  $\int_0^3 f(x) dx = 6$  and  $\int_3^7 f(x) dx = 9$ , find  $\int_0^7 f(x) dx$ .
- a. 13
  - b. 3
  - c. 15
  - d. 54

- 73 Given  $\int_0^5 f(x) dx = 14$ , find  $\int_5^0 f(x) dx$ .
- a. 14
  - b.  $\frac{1}{14}$
  - c. 0
  - d. -14
- 74 Given  $\int_1^5 f(x) dx = 20$  and  $\int_1^5 3 dx = 12$ , find  $\int_1^5 [f(x) - 3] dx$ .
- a. -8
  - b. 8
  - c. 32
  - d. 17
- 75 Given  $\int_0^4 f(x) dx = 9$  and  $\int_0^4 g(x) dx = 5$ , find  $\int_0^4 [f(x) + g(x)] dx$ .
- a.  $\frac{9}{5}$
  - b. 45
  - c. 4
  - d. 14

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### MCQ — Finding a function from its derivative

- 76 A function satisfies  $f'(x) = 8x^3 - 6$  and  $f(1) = 3$ . Find  $f(x)$ .
- a.  $2x^4 - 6x + 7$
  - b.  $2x^4 - 6x + 3$
  - c.  $8x^4 - 6x + 7$
  - d.  $2x^4 - 6x - 7$

77 A function satisfies  $f'(x) = 4x^2 + 3x$  and  $f(0) = 2$ . Find  $f(x)$ .

- a.  $\frac{4x^3}{3} + 3x^2 + 2$
- b.  $\frac{4x^3}{3} + 1.5x^2$
- c.  $\frac{4x^3}{3} + 1.5x^2 + 2$
- d.  $4x^3 + 1.5x^2 + 2$

78 A function satisfies  $f'(x) = 6x - 5$  and  $f(3) = 20$ . Find  $f(x)$ .

- a.  $3x^2 - 5x + 20$
- b.  $3x^2 - 5x + 8$
- c.  $6x^2 - 5x + 8$
- d.  $3x^2 - 5x - 8$

79 A function satisfies  $f'(x) = 2x + 7$  and  $f(-1) = 4$ . Find  $f(x)$ .

- a.  $x^2 + 7x + 10$
- b.  $2x^2 + 7x + 10$
- c.  $x^2 + 7x - 10$
- d.  $x^2 + 7x + 4$

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### MCQ — Average value of a function

80 Find the average value of  $f(x) = x^2$  on  $[0, 4]$ , using  $\text{avg} = \frac{1}{b-a} \int_a^b f(x) dx$ .

- a.  $\frac{64}{3}$
- b. 16
- c.  $\frac{16}{3}$
- d. 4

81 Find the average value of  $f(x) = 3x + 2$  on  $[0, 4]$ .

- a. 32
- b. 4
- c. 14
- d. 8

82 Find the average value of  $f(x) = x^2 - 1$  on  $[1, 3]$ .

- a.  $\frac{8}{3}$
- b.  $\frac{10}{3}$
- c.  $\frac{20}{3}$
- d. 6

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### MCQ — Word problems: integration in context

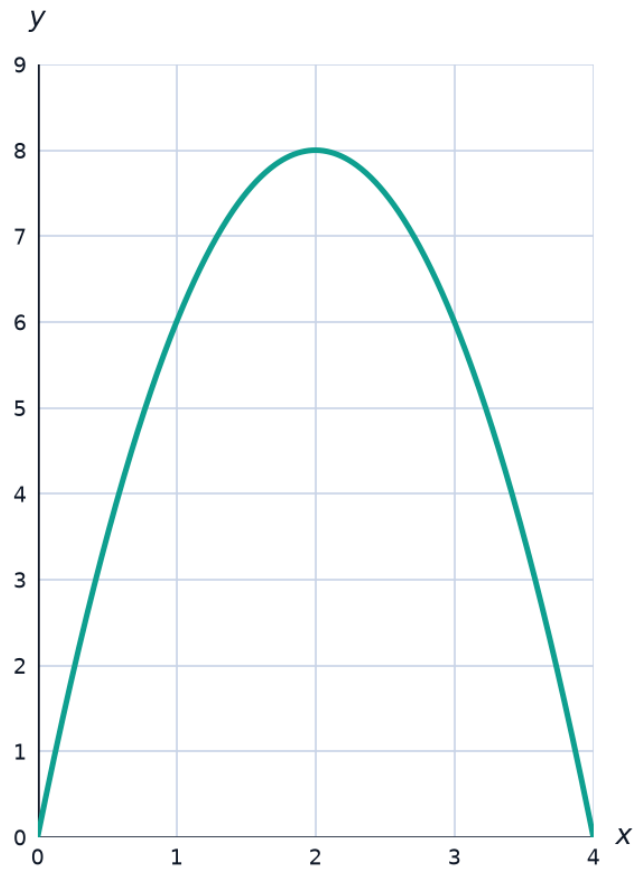
83 A particle moves along a line with velocity  $v(t) = 8t - 2t^2$  (m/s) for  $0 \leq t \leq 4$ . Find the particle's displacement over this interval.

- a. 64
- b.  $\frac{64}{3}$
- c. 32
- d.  $\frac{192}{3}$

84 Water flows into a tank at a rate of  $r(t) = 30 - 3t$  liters per minute, for  $0 \leq t \leq 10$ . Find the total volume of water that flows in during this time.

- a. 300
- b. 150
- c. 450
- d. 100

- 85 The graph shows a particle's velocity  $v(t) = 8t - 2t^2$  for  $0 \leq t \leq 4$ . Find the particle's displacement (the area under the curve).



- a.  $\frac{64}{3}$
- b. 16
- c. 32
- d. 64
- 86 A factory's production rate is  $p(t) = 50 + 4t$  items per hour, for  $0 \leq t \leq 8$ . Find the total number of items produced during this 8-hour period.
- a. 528
- b. 656
- c. 450
- d. 400